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SEMESTER 1 EXAMINATIONS 2024/2025

MODULE: CSC1048 - Computability and Complexity

PROGRAMME(S):

CASE	BSc Computer Apps (Sft.Eng.)
COMSCI	BSc Computer Science
ECGTE	Engineering & Computing GTE
ECSA	Study Abroad Program EC
ECSAO	Study Abroad Progrgam EC

YEAR OF STUDY: 1,3

EXAMINERS: Dr. David Sinclair (Internal) (Ext. 5510)

TIME ALLOWED: 2 hours

INSTRUCTIONS: Answer 4 Questions.
All questions carry equal marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO

The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

Requirements for this paper (Please mark (X) as appropriate)

☐	*Log Tables*	☐	*Thermodynamic Tables*
☐	*Graph Paper*	☐	*Actuarial Tables*
☐	*Dictionaries*	☐	*MCQ Only - Do not publish*
☐	*Statistical Tables*	☐	*Attached Answer Sheet*

Section A

QUESTION 1 ***[Total marks: 25]***

1(a) [5 Marks]

Using examples, show how the `let` and `where` constructs are used in Haskell.

1(b) [5 Marks]

Using an example, illustrate how the `case` statement can be used in conjunction with pattern matching in Haskell.

1(c) [15 Marks]

Representing sets as lists, write Haskell functions that implements set union and set difference. Remember sets do not have duplicates. The function should use class constraints to ensure that it will work for sets of integers, floats, and characters.

[End Question 1]

QUESTION 2 ***[Total marks: 25]***

2(a) [5 Marks]

In the context of Haskell, describe what is meant by *lazy evaluation* and give an example of it.

2(b) [5 Marks]

Write a Haskell *data type* for a binary tree.

2(c) [15 Marks]

A reverse *monkey puzzle sort* of a list is performed by storing the elements of the list into a binary tree and then traversing the tree so that the nodes of a subtree are visited in the order right child, parent and then left child. Write a reverse monkey puzzle sort in Haskell.

[End Question 2]

Section B

QUESTION 3 ***[Total marks: 25]***

3(a) [5 Marks]

Formally define a *context-free grammar*.

3(b) [10 Marks]

Formally describe the simplest automaton that can recognise a context-free grammar.

3(c) [10 Marks]

Construct an automaton (of the type indicated in part (b)) that recognises the language $L = \{ww^r | w^r$ is the reverse of the string $w\}$ over the alphabet $\Sigma = \{a, b, c\}$.

[End Question 3]

QUESTION 4 ***[Total marks: 25]***

4(a) [5 Marks]

In *computability* what does it mean when a problem is *undecidable*?

4(b) [13 Marks]

Prove that the *Halting Problem* is undecidable.

4(c) [7 Marks]

Describe how *reducibility* can be used to show a problem is undecidable.

[End Question 4]

QUESTION 5 ***[Total marks: 25]***

5(a) [5 Marks]

Focusing on the *transition function*, describe the difference between a *deterministic finite automaton* and a *nondeterministic automaton*.

5(b) [10 Marks]

Compare and contrast a *Linear Bounded Automaton*, a *Pushdown Automaton* and a *Turing Machine*.

5(c) [10 Marks]

Design a *Turing Machine* that takes 2 unary numbers on the tape, where the first number is greater than the second number, and computes and writes a unary number to the tape that is the first number minus the second number.

[End Question 5]

[END OF EXAM]